15. Suppose a 3×5 matrix A has three pivot columns. Is $\text{Col}\,A=\mathbb{R}^3$? Is $\text{Nul}\,A=\mathbb{R}^2$? Explain your answers.

Solution: By Theorem 13, the 3 pivot columns of A form a basis for $\text{Col}\,A$.

By defn, a basis for $\text{Col}\,A$ forms a set of 3 3×1 linearly independent vectors which span $\text{Col}\,A$. Let the columns be b_1, b_2, b_3.

Let B be a 3×3 matrix where $B=[b_1\ b_2\ b_3]$.

By the Invertible Matrix Theorem and the equivalence of the statements (e) and (h), we know that the columns of B span $\mathbb{R}^3$.

$\therefore \text{Col}\,A=\mathbb{R}^3$

The nullspace of A must be a subspace of $\mathbb{R}^5$, according to Theorem 12. From the defn. of subspace, we know it must also be a set in $\mathbb{R}^5$.

$\therefore \text{Nul}\,A\neq\mathbb{R}^2$ II

16. Suppose a 4×7 matrix A has three pivot columns. Is $\text{Col}\,A=\mathbb{R}^3$? What is the dimension of $\text{Nul}\,A$? Explain your answers.

Solution: By Theorem 13, the 3 pivot columns of A form a basis for $\text{Col}\,A$.

By defn, a basis for $\text{Col}\,A$ forms a set of 3 4×1 linearly independent vectors which span $\text{Col}\,A$.

$\therefore$ The given vectors are of dimension 4×1, $\text{Col}\,A\neq\mathbb{R}^3$.

A has 4 free variables. $\therefore$ From Example 2, the dimension of $\text{Nul}\,A$ must be 4. II

In Exercises 17 and 18, mark each statement True or False. Justify each answer. Here A is an $m\times n$ matrix.

17) a) If $B=\{v_1,\dots,v_p\}$ is a basis for a subspace H and if $x=c_1v_1+\dots+c_pv_p$, then $c_1,\dots,c_p$ are the coordinates of x relative to the basis B.

Soln: This statement is True by the defn. of coordinate vector given on Page 156.

b) Each line in $\mathbb{R}^n$ is a one-dimensional subspace of $\mathbb{R}^n$.